# Biomedical Knowledge Graph Construction Using Large Language Models: Evaluating Generalisation Beyond Benchmark Datasets

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# Introduction

# Chapter 2

# Literature Review

### 2.0.1 Definition

Biomedical Information Extraction (IE) is the process of automatically identifying structured information - entities such as genes, diseases, chemicals and species, and the relationships between them - from unstructured biomedical text such as journal abstracts and full-text articles. In this project, IE is treated as the first stage of a larger system: extracted entities and relations are later assembled into a knowledge graph and validated against biomedical ontologies **[REFERENCE]** (section)

### 2.0.2 Motivation

The volume of biomedical literature has expanded at a rate that exceeds the capacity of manual review, with previous noncurrent analyses demonstrating an exponential growth in the number of biomedical publications [1]. This growth is supported by more recent data - a mapping of over 20 million PubMed abstracts has demonstrated the magnitude of contemporary biomedical literature to the level that the volume of biomedical and life-science publications has made it difficult for researchers to keep track of new literature [2]. This is not only a volume issue, but also an issue with discovery - log analysis of PubMed search behaviour shows that over 80% of views fall within the first page (top 20 results) of returned literature [3], meaning a substantial majority of literature goes largely unseen by researchers relying on manual review of ranked results.

Together, these findings motivate the need for automated IE for two reasons: the breadth of literature is growing faster than it can be analysed [1, 2] and that there is a significant attention bottleneck that automated extraction and structuring could help mitigate [3]. This motivation supports the choice of PubMed as the source of the contemporary corpus for this thesis, as a corpus that is both large and continuously updated is exactly the setting in which static benchmark-trained pipelines are most likely to be used.

### 2.0.3 Tasks

Biomedical IE is usually comprised of several sub-tasks: Named Entity Recognition (NER), which identifies and stores types of spans of text referring to biomedical concepts, and Relation Extraction (RE), which identifies semantic relationships between recognised entities and Entity Normalisation (EN), which maps extracted mentions to pre-defined identifiers from an existing ontology [4, 5]. Long or complex sentences, which are common in biomedical abstracts, summarising multiple findings into a single sentence, can cause a significant challenge for these tasks [6].

### 2.0.4 Challenges

Beyond task complexity, biomedical text includes domain-specific challenges that separate it from general IE: dense and evolving terminology [7, 8], high rates of entity name ambiguity [9] (e.g.: synonymy such as "cerebral hemorrhage" and "intra-cerebral micro-bleeds" or, less-commonly, polysemy), and frequent use of abbreviations, many of which are ambiguous, defined within that paper, or polysemous.

### 2.0.5 Traditional Methods

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# Chapter 3

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